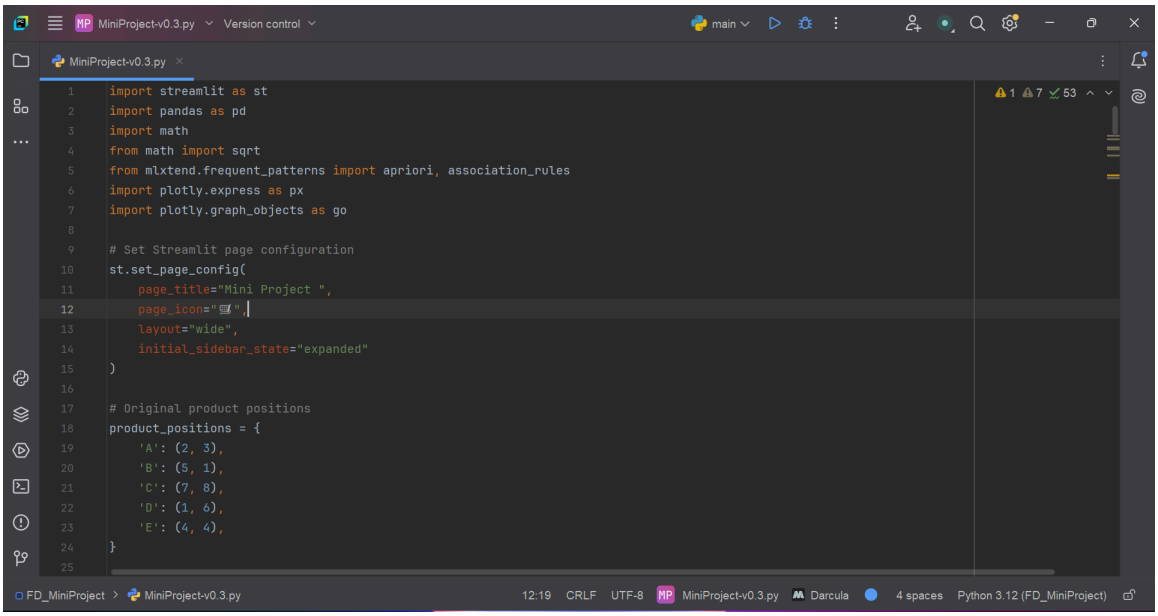
Market Analyzer - Mini Report

1. Project Overview

Objective: Analyze purchasing patterns using the Apriori algorithm, optimize store layout based on frequent itemsets, and calculate customer path distances to improve efficiency.

2.Tools and Technologies:

- Python
- Streamlit (web app for UI)
- Pandas, mlxtend (Apriori), Plotly (visualizations)



3.Dataset Structure

The dataset used for this project was loaded from an **external file** in either **Excel (`.xlsx`)** or **CSV (`.csv`)** format using Streamlit's file uploader component. It follows a typical structure for Market Basket Analysis:

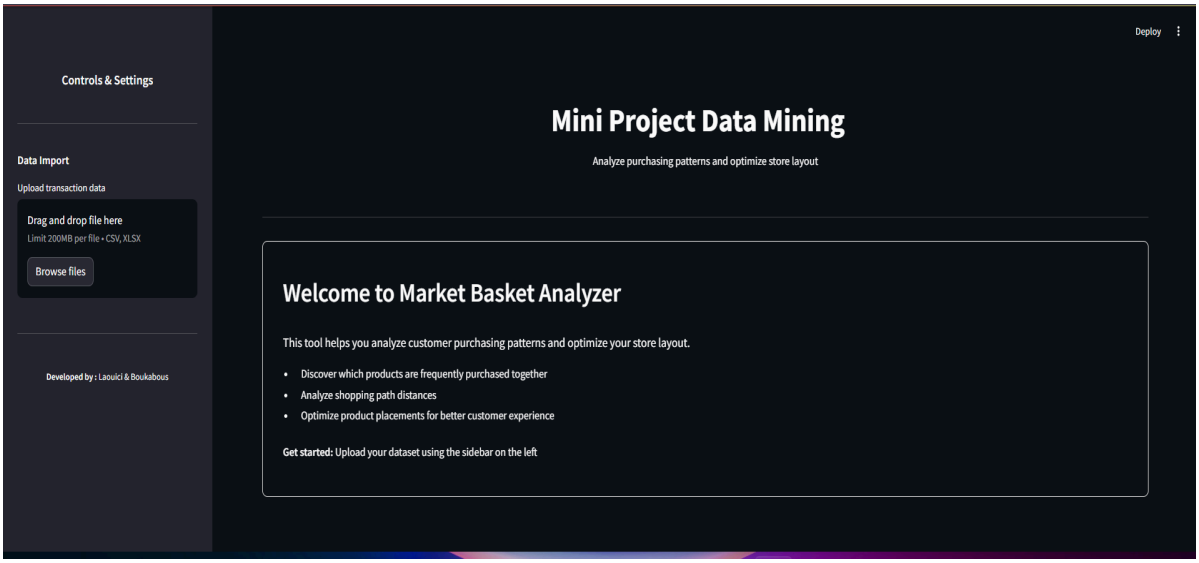
- `transaction_ID`: A unique identifier for each transaction.
- **Product Columns** (A, B, C, etc.): Each column represents a product.
  - The values are **binary**:
    - 1 → The product was purchased in that transaction.
    - 0 → The product was not purchased.

Overview :

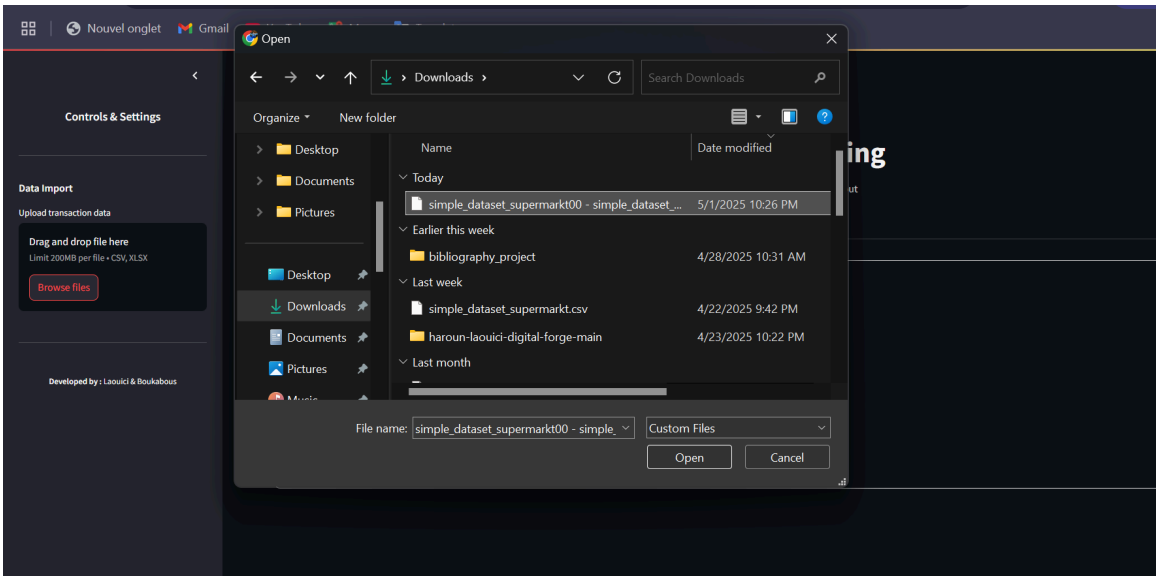
	A (1)	B (2)	C (3)	D (4)	E (5)
1	A	B	C	D	E
2	0	1	1	0	0
3	1	1	1	0	1
4	1	1	0	1	1
5	1	0	1	0	1
6	1	1	0	0	0
7	0	1	1	0	0
8	0	0	1	0	1
9	0	1	0	1	0
10	1	0	1	0	0

App Interface Overview – Home Page

The home interface of the application is designed to be **clean, intuitive, and user-friendly**, built using **Streamlit**. Upon launching the app, the user is greeted with a sidebar and a main content area.



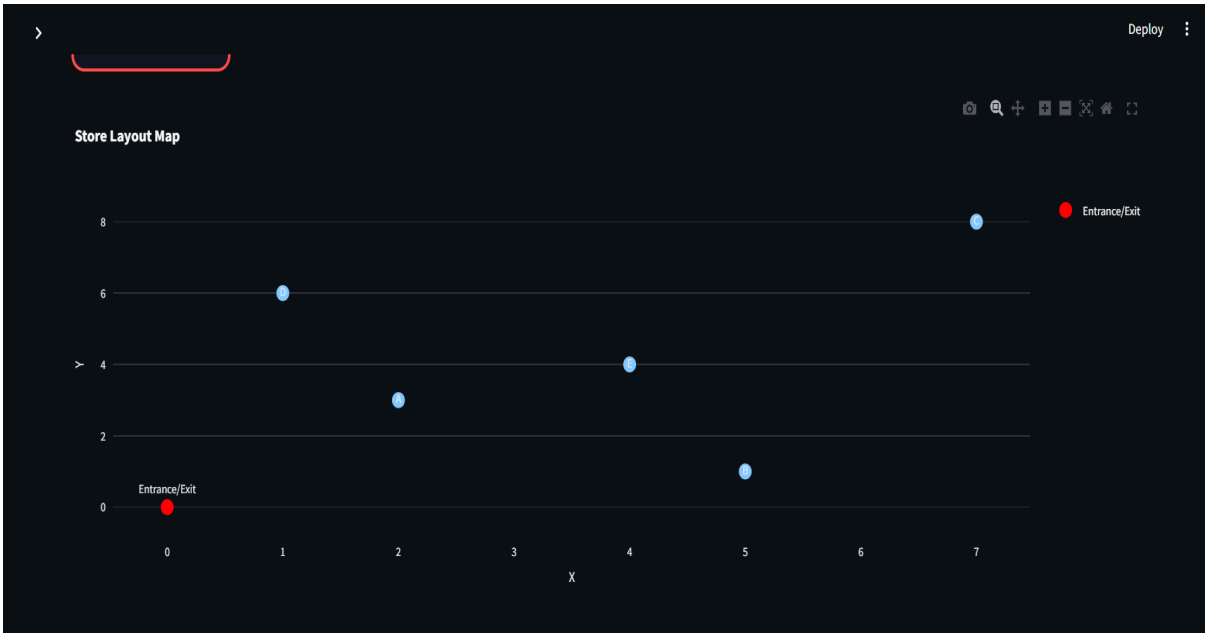
Uploading the csv file :



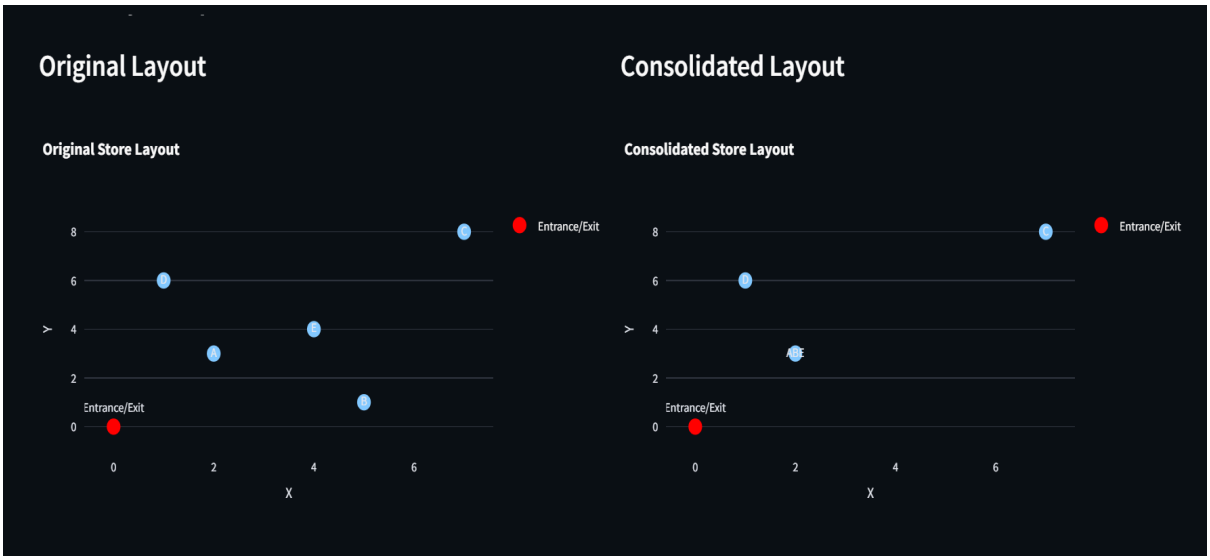
Store Layout Map:

The **Store Layout Map** visually represents the physical arrangement of products inside a retail store.

- **Each blue point** on the map corresponds to a **product**.
- The product's **coordinates (x, y)** reflect its **position** in the store grid.
- This layout helps simulate and analyze customer navigation through the store.
- The **entrance** is considered at point **(0, 0) (the red point)**, and distance metrics are calculated based on how far products are placed from this point.



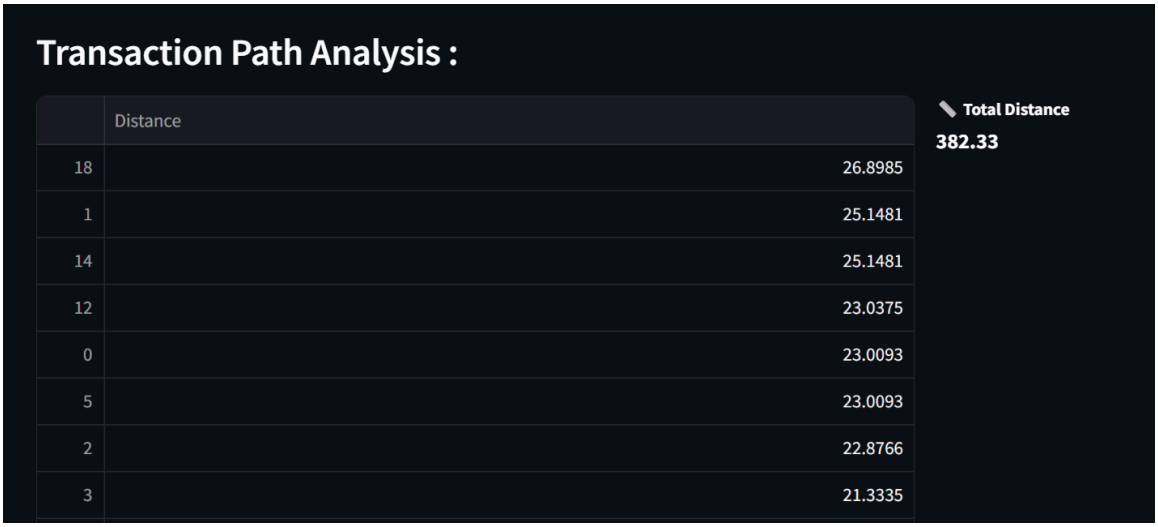
Original Layout VS Consolidated Layout:



Total distance :

Before:

in this case the total distance before was **382.33**



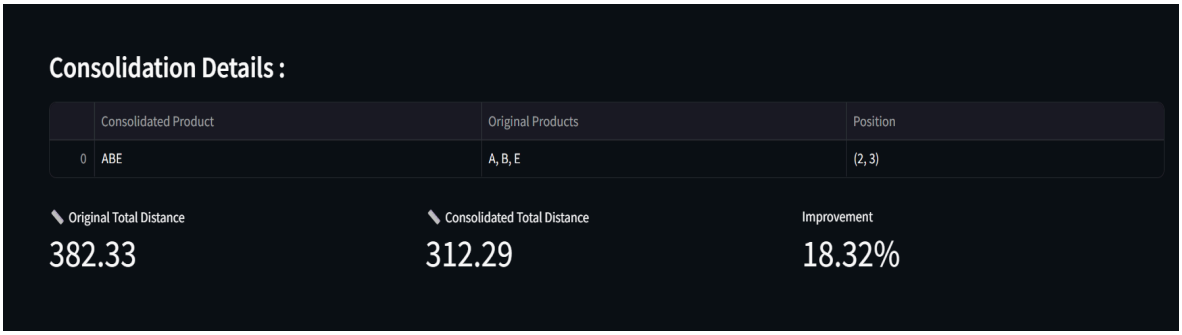
Transaction Path Analysis :

	Distance
18	26.8985
1	25.1481
14	25.1481
12	23.0375
0	23.0093
5	23.0093
2	22.8766
3	21.3335

Total Distance
382.33

After:

After consolidation the total distance has been reduced to **312.29** **as the you can see down here** ⬇️



Consolidation Details :

	Consolidated Product	Original Products	Position
0	ABE	A, B, E	(2, 3)

Original Total Distance
382.33

Consolidated Total Distance
312.29

Improvement
18.32%